

Amazon Review Sentiment Analysis

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Abstract

Online reviews contain more detail than star ratings, but the relationship between the written text and the selected rating is not always clear. This study compares sentiment and language in Amazon reviews from the Beauty & Personal Care and Sports & Outdoors categories between 2015 and September 2023. We use a balanced sample of 189,587 reviews and apply VADER to measure sentiment. We compare sentiment distributions, analyse changes over time, measure agreement between VADER and star-rating-based labels, and identify distinctive words and phrases using weighted log-odds.

The results show that the two categories have very similar overall sentiment distributions and follow broadly similar patterns over time. VADER has moderate agreement with star ratings and performs well for positive reviews, but it has difficulty with neutral and longer reviews. The vocabulary analysis shows that positive reviews use similar general language, while negative reviews contain more category-specific complaints. These findings show that overall sentiment scores can hide important differences in how customers describe their product experiences.

CCS Concepts

• **Information systems** → **Sentiment analysis**; • **Computing methodologies** → *Natural language processing*.

Keywords

sentiment analysis, Amazon reviews, online consumer reviews, VADER, star ratings, temporal analysis

ACM Reference Format:

Elena Dumitrache (5812852), Andrei Nita (5775523), and Cosmin Visan (5710820). 2026. Amazon Review Sentiment Analysis. In *DSAIT4055: Web Science and Engineering*, TU Delft. ACM, New York, NY, USA, 8 pages.

1 Introduction

Online product reviews are an important source of information on the Web. They allow customers to describe their experiences and help other people decide which products to buy. Most review platforms also use star ratings as a short summary of customer satisfaction. However, a star rating does not always fully represent what is written in the review. A customer may give a high rating while mentioning a problem, or give a low rating while still describing some positive features.

The way people write reviews may also depend on the type of product. In this study, we compare reviews from the Beauty & Personal Care and Sports & Outdoors categories. These categories were chosen because they represent different product experiences while still using the same Amazon review system. Beauty reviews often discuss properties such as scent, appearance, comfort, and

effectiveness. Sports reviews are more likely to discuss fit, material quality, durability, and performance. This comparison allows us to study if different product categories have similar overall sentiment while using different language to describe their experiences.

This paper is a 'real' research paper with an analytical focus. Its goal is to gain a better understanding of consumer sentiment in online reviews rather than to develop a new technical system. We use reviews from the Amazon Reviews 2023 dataset [4], covering the period from 2015 to September 2023. A balanced sample is created for each category and year so that the two categories can be compared fairly.

We apply VADER [5] to measure the sentiment of each review. We then compare the sentiment distributions of the two categories, study how sentiment changes over time, and examine how well VADER predictions agree with the star ratings given by reviewers. Finally, we use weighted log-odds to identify words and phrases that are especially common in high-rated and low-rated reviews.

Based on these goals, we answer the following research questions:

- RQ1** How does the distribution of VADER sentiment scores differ between Beauty & Personal Care and Sports & Outdoors reviews?
- RQ2** How does average review sentiment change between 2015 and September 2023, and do the two categories follow different temporal patterns?
- RQ3** How well do VADER sentiment predictions agree with star-rating-based reference labels, and how does this agreement differ by category, review length, and sentiment class?
- RQ4** Which words and phrases distinguish high-rated from low-rated reviews within each category, and which terms distinguish the two categories from each other?

The results show that the two categories have very similar overall sentiment distributions and follow broadly similar changes over time. VADER has a moderate relationship with star ratings and performs well for positive reviews, but it has difficulty identifying neutral reviews. The vocabulary analysis also shows that positive reviews often use similar general words, while negative reviews contain clearer category-specific complaints. These findings suggest that similar overall sentiment can hide important differences in how customers describe their product experiences.

2 Background and Related Work

Online reviews provide more detail than star ratings because they explain why a customer liked or disliked a product. Mudambi and Schuff [6] showed that the usefulness of an Amazon review can depend on its length, rating, and product type. This suggests that the way opinions are written may differ between product categories.

Sentiment analysis is used to identify positive, negative, or neutral opinions in text. Common approaches include lexicon-based methods, traditional machine-learning models, and deep-learning

models. Qorich and El Ouazzani [8] and Ashbaugh and Zhang [2] applied machine-learning and deep-learning methods to customer reviews. These methods can achieve strong results, but they require labelled training data and more computational resources. Qi and Shabrina [7] also showed that the performance of sentiment-analysis methods depends on the dataset and the goal of the study.

In this paper, we use VADER, a lexicon- and rule-based sentiment-analysis tool developed by Hutto and Gilbert [5]. VADER assigns positive, negative, neutral, and compound sentiment scores to a text. It also considers features such as negation, punctuation, capitalisation, and words that strengthen or weaken sentiment. Since it does not require training data, it is suitable for analysing a large number of reviews. However, VADER was mainly designed for short and informal social-media text. Product reviews can be longer, include mixed opinions, and contain words that are specific to a product category.

Star ratings and written reviews also do not always express the same opinion. For example, a customer may give a high rating while mentioning a problem. Al-Natour and Turetken [1] found that agreement between sentiment-analysis tools and star ratings can depend on product type and review length. For this reason, we use star ratings as reference labels rather than treating them as perfect ground-truth sentiment labels.

Previous studies have mainly focused on improving sentiment classification or comparing review text with star ratings. Other research has shown that sentiment vocabulary can differ between product domains [3]. Our study combines these ideas by comparing sentiment distributions, changes over time, agreement with star ratings, and distinctive vocabulary across two Amazon product categories. While Al-Natour and Turetken compared sentiment-analysis tools with star ratings, our study focuses on a recent large-scale Amazon dataset and examines how agreement differs across product categories, review lengths, and sentiment classes.

3 Dataset

We use the Amazon Reviews 2023 dataset provided by the McAuley Lab through the UCSD Recommender Systems Datasets page and Hugging Face [4]. The full dataset contains more than 570 million reviews from 33 Amazon product categories. Each review includes information such as the written title and text, star rating, timestamp, number of helpful votes, and verified-purchase status.

For this study, we use reviews from the Beauty & Personal Care and Sports & Outdoors categories. We include reviews published between 2015 and September 2023. The year 2023 is incomplete because the dataset only contains reviews up to September.

To make the categories and years comparable, we created a balanced sample containing 11,112 reviews for each category and year. This produced an initial sample of 200,016 reviews across 18 category-year groups. The sampling was performed randomly with a fixed seed of 42. Giving each year the same sample size prevents years with more reviews from having a larger influence on the results.

After removing duplicate reviews, non-English reviews, and unusable text, the final dataset contained 189,587 reviews. Of these, 94,261 belong to Beauty & Personal Care and 95,326 to Sports & Outdoors. The review title and main review text were combined and

used as the text input for the sentiment and vocabulary analyses. Star ratings and timestamps were used for the rating-agreement and temporal analyses.

4 Methodology

The study was implemented as a seven-stage Python pipeline. The first three stages prepare the data by sampling the reviews, cleaning the text, and calculating VADER sentiment scores. The remaining four stages correspond to the four research questions: comparing sentiment distributions, analysing changes over time, measuring agreement between VADER and star ratings, and identifying distinctive vocabulary. Each stage reads the output of the previous stage and saves its own results, allowing separate parts of the analysis to be repeated without running the complete pipeline. The main settings, including the sample size, random seed, VADER thresholds, and bootstrap parameters, are stored in a separate configuration file.

4.1 Sampling

The original Amazon dataset is too large to load fully into memory. Therefore, the reviews were read as a stream from Hugging Face. For each product category and year, the code used reservoir sampling to select a random sample while processing the data only once.

A separate reservoir was created for every category-year combination. Each of them kept at most 11,112 reviews. When a new review was read, it either filled an empty place in it or randomly replaced an existing review. This method gives every review in the same category and year an equal chance of being selected.

4.2 Text Preprocessing

The review title and main text were cleaned separately. HTML tags, HTML character codes, Amazon product markers, and repeated spaces were removed. Different apostrophe symbols were also changed to a standard apostrophe. This keeps contractions such as “don’t” together during the later word analysis.

After cleaning, the title and review body were combined into one field called `review_text`. If a title was present, it was placed before the main review text. The code also recorded the character length and word count of every review. The short ones were kept because their length is later used when studying the agreement between VADER and star ratings.

Reviews containing no usable text were removed. Exact duplicate texts were also removed, but only when they appeared in the same product category and year. This avoids counting copied reviews more than once while allowing the same short phrase to appear naturally in different years or categories.

A two-step language filter was used to keep English reviews. First, reviews containing less than 85% ASCII characters were removed. The remaining reviews were checked using the `langdetect` library, and only reviews classified as English were kept. The language detector also used the fixed seed of 42.

4.3 Sentiment Scoring

Each cleaned review was scored using VADER [5]. VADER returns four values: positive, negative, neutral, and compound sentiment.

The compound score combines the sentiment of the complete review into a value between -1 and 1 .

The standard VADER thresholds were used to create sentiment labels. Reviews with a compound score greater than or equal to 0.05 were labelled positive. Reviews with a score less than or equal to -0.05 were labelled negative. Scores between these values were labelled neutral.

Star ratings were converted into a second set of sentiment labels. Ratings of four or five stars were labelled positive, three-star ratings were labelled neutral, and one- or two-star ratings were labelled negative. These labels were used as reference labels when comparing star ratings with VADER. They were not treated as perfect ground truth.

4.4 Statistical Methods

To compare the sentiment distributions of the two categories, the code calculated descriptive statistics and created density, violin, and empirical cumulative distribution plots. The Mann–Whitney U test was used to compare the positions of the distributions, while the Kolmogorov–Smirnov test was used to compare their overall shapes. Cliff’s delta and rank-biserial correlation were calculated to measure the size of the difference. Confidence intervals for means, medians, and their differences were estimated using bootstrap sampling.

For the temporal analysis, the mean VADER compound score was calculated for each category and year. Bootstrap confidence intervals were added to the yearly means. The Mann–Kendall test and Sen’s slope were used to check for monotonic trends. Review-level linear regression was also applied separately to each category. An additional regression model included an interaction between year and category to test if the categories changed differently over time. Yearly category comparisons used Mann–Whitney U tests, followed by Holm–Bonferroni correction for multiple comparisons.

Agreement between VADER and star ratings was measured using Spearman correlation, accuracy, macro-F1, precision, recall, confusion matrices, and quadratic-weighted Cohen’s kappa. Accuracy was also compared with a baseline that always predicts the most common rating class. Agreement was calculated separately for each category and for reviews in four length groups: at most 5 words, 6–20 words, 21–50 words, and more than 50 words. The code also selected examples of strong disagreement, including high-rated reviews with strongly negative VADER scores and low-rated reviews with strongly positive scores.

4.5 Vocabulary Analysis

Distinctive words and two-word phrases were identified using weighted log-odds with an informative prior. Three comparisons were performed: high-rated against low-rated Beauty reviews, high-rated against low-rated Sports reviews, and all Beauty reviews against all Sports reviews.

The vocabulary included single words and two-word phrases that appeared at least ten times. Common English stopwords were removed, but the negation words “no”, “not”, and “nor” were kept because they can change the meaning of a sentence. Contractions such as “don’t” and “wasn’t” were also kept as complete tokens.

Star-rating labels were used to define the high-rated and low-rated groups. This keeps the vocabulary analysis independent from VADER. TF–IDF scores were also calculated as a secondary check of the main terms identified by the weighted log-odds method.

All bootstrap procedures used 10,000 repetitions and a 95% confidence level. Intermediate datasets were stored in Parquet format, while statistical results and tables were saved as JSON, CSV, and LaTeX files. Figures were generated in both PNG and PDF formats.

5 Results

The results are presented in the order of the four research questions. In RQ3, star ratings are used to compare with VADER predictions, while in RQ4 they are used to separate high-rated and low-rated reviews.

5.1 RQ1: Sentiment Distributions

Figure 1 shows where the VADER compound scores are most concentrated. Scores closer to 1 represent more positive reviews, while scores closer to -1 represent more negative reviews. Both categories have a large peak near the positive end of the scale. The two curves overlap almost completely, showing that Beauty and Sports reviews have very similar sentiment distributions.

If one curve had been clearly shifted towards the right, that category would have contained more positive reviews. If it had been shifted towards the left, it would have contained more negative reviews. Neither pattern is clearly visible.

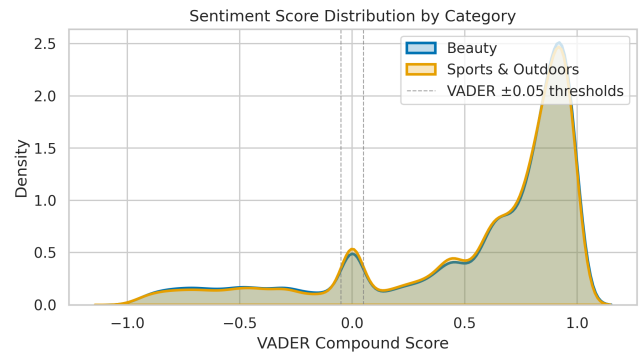


Figure 1: Density of VADER compound scores for Beauty and Sports reviews. The dashed lines show the thresholds used to separate negative, neutral, and positive sentiment.

Figure 2 presents the same distributions in another form. Wider parts of each shape represent score ranges containing more reviews. Both shapes are widest near the positive end and have long, narrow tails towards negative scores. This again shows that most reviews are positive, while strongly negative reviews are less common.

The central marker represents the median score. Beauty had a slightly higher median than Sports, with 0.785 compared with 0.777 . However, this difference is very small compared with the complete range of sentiment scores.

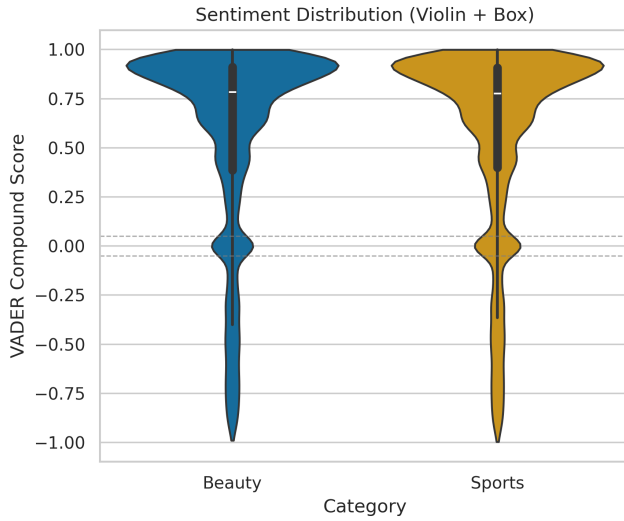


Figure 2: Violin plots showing the shape and central values of the sentiment distributions. Wider areas contain more reviews.

Figure 3 shows the cumulative distributions. At each sentiment score, the vertical axis gives the percentage of reviews with that score or a lower score. The two lines remain close across almost the complete scale.

The largest distance between the lines was approximately 0.016. Therefore, even at the point where the categories differed the most, their cumulative percentages differed by only about 1.6 percentage points. If the lines had been far apart, this would have shown a clearer difference between the categories.

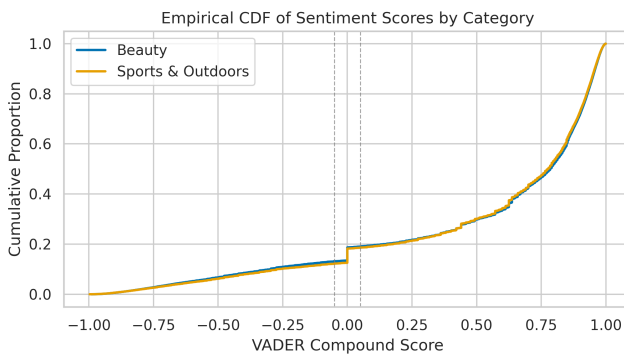


Figure 3: Empirical cumulative distribution of VADER compound scores. The close lines indicate that the two categories have similar distributions.

Beauty had a mean compound score of 0.556, while Sports had a mean of 0.558. Around 81% of the reviews in both categories were classified as positive. The statistical tests found a difference between the distributions, but Cliff's delta was only 0.0095. The difference can therefore be detected because the sample is large, but its actual size is very small.

5.2 RQ2: Sentiment over Time

Figure 4 shows the average compound score for each category and year. A higher point represents more positive average sentiment. The shaded areas around the lines show 95% confidence intervals.

The two categories follow a broadly similar pattern. Beauty started with higher values in 2015 and 2016, after which its average sentiment decreased. Both categories reached their highest average value in 2019. Sentiment then became lower after 2020. Beauty reached its lowest point in 2021, while Sports reached its lowest point in 2022. Both categories increased again in 2023, although the data for that year only include reviews until September.

If sentiment had continuously improved, the lines would have moved mainly upwards. If sentiment had continuously decreased, the lines would have moved mainly downwards. Instead, both lines rise and fall, showing that the changes were not continuous in one direction.

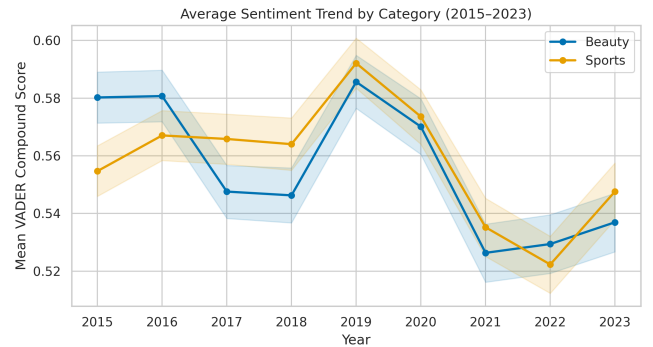


Figure 4: Average VADER compound score by category and year. Shaded areas show 95% bootstrap confidence intervals. The 2023 data cover January to September.

The Mann–Kendall tests found no clear monotonic trend for either Beauty ($p = 0.118$) or Sports ($p = 0.348$). The regression analysis found a slightly stronger decline for Beauty, but the model explained less than 0.1% of the total variation in sentiment. When comparing the categories year by year, a significant difference was found only in 2015 and 2016.

5.3 RQ3: Agreement with Star Ratings

VADER compound scores had a moderate positive relationship with star ratings, with a Spearman correlation of $\rho = 0.510$. Agreement between the VADER labels and rating-based labels reached an accuracy of 80.9%, a macro-F1 score of 0.542, and a quadratic-weighted kappa of 0.597.

Positive reviews formed the largest class in the dataset. A simple method that always predicted positive would already reach an accuracy of 77.0%. The confusion matrices and the results for each sentiment class are therefore important for understanding where VADER performed well and where it had more difficulty.

Figure 5 shows agreement for different review lengths. The height of each bar represents the percentage of reviews for which VADER and the star-rating label produced the same sentiment class.

Agreement was highest for reviews containing 6–20 words, at 82.7%. It decreased to 80.6% for reviews containing 21–50 words and to 78.0% for reviews longer than 50 words. Very short reviews performed similarly to the 6–20 word group.

If all bars had approximately the same height, review length would not appear to be related to agreement. Instead, the lower bars for longer reviews show that agreement becomes weaker as reviews contain more text.

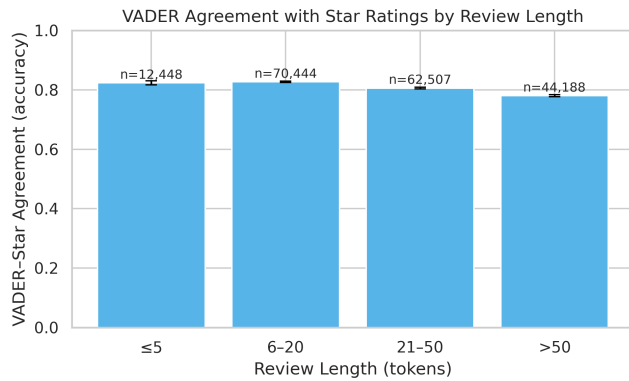


Figure 5: Agreement between VADER and rating-based labels for different review-length groups. The error bars show 95% confidence intervals.

Figure 6 shows the confusion matrix for Beauty reviews. The rows represent the labels created from star ratings, while the columns represent the VADER predictions. Cells on the main diagonal represent cases where the two labels agree.

The largest cell is the positive-positive cell. This means that VADER correctly identified most high-rated Beauty reviews as positive. The neutral-neutral cell is much smaller. Most three-star Beauty reviews were classified as positive rather than neutral.

If VADER had performed equally well for all sentiment classes, all three diagonal cells would have been clearly dominant. Instead, the figure shows that the high overall accuracy mainly comes from the positive class.

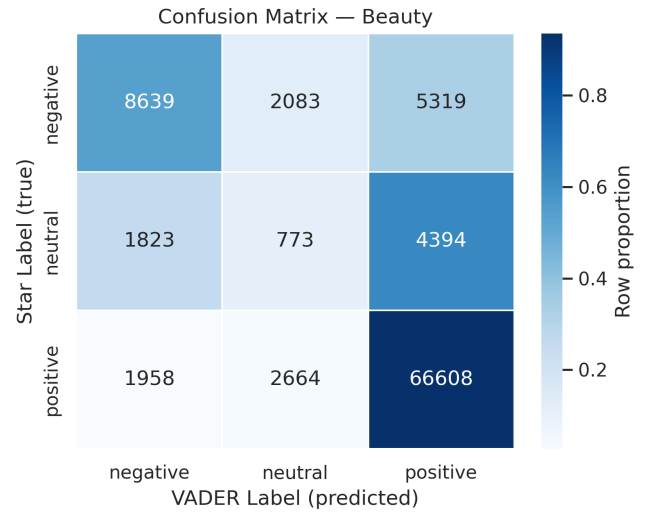


Figure 6: Confusion matrix for Beauty reviews. Rows show rating-based labels and columns show VADER predictions.

The Sports confusion matrix in Figure 7 shows almost the same pattern. Positive reviews are recognised well, negative reviews are recognised less consistently, and neutral reviews are usually classified as positive.

The similarity between the two confusion matrices shows that VADER behaves similarly across the two categories. Sports had slightly higher exact agreement, while Beauty had slightly higher correlation and weighted kappa. Therefore, neither category performed better across every measure.

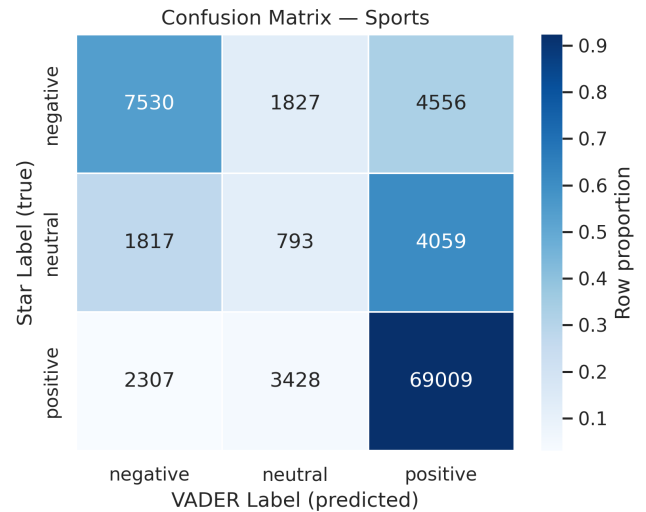


Figure 7: Confusion matrix for Sports reviews. Rows show rating-based labels and columns show VADER predictions.

The class-level results support what is visible in the confusion matrices. The F1 score was 0.904 for positive reviews, 0.599 for negative reviews, and only 0.124 for neutral reviews. VADER therefore

performed well for positive reviews but had difficulty separating neutral reviews from positive ones.

5.4 RQ4: Distinctive Vocabulary

The full vocabulary plots for RQ4 are presented in Appendix A, Figure 8. They show the words and two-word phrases that best distinguish the different review groups. The length of each bar represents how strongly a term is connected to one group compared with the other. It does not simply represent how often the term appears.

In Figures 8a and 8b, bars extending to the right are more strongly linked to high-rated reviews, while bars extending to the left are more strongly linked to low-rated reviews. In Figure 8c, the direction represents the product category rather than positive or negative sentiment.

The high-rated sides of both categories contain similar general words, including *great*, *love*, *perfect*, and *easy*. High-rated Sports reviews also contain descriptions related to product use, such as *comfortable*, *fits*, *sturdy*, and *good quality*.

Low-rated Beauty reviews contain expressions such as *doesn't work*, *waste money*, *disappointed*, *return*, and *not worth*. These expressions mainly describe products that did not produce the expected result.

Low-rated Sports reviews contain more words related to physical damage and product quality, including *broke*, *broken*, *poor quality*, *cheap*, *useless*, and *junk*. The strong position of *not* on the negative side of both plots also shows why negation words were kept during text preprocessing.

The direct category comparison in Figure 8c shows clear differences in product-related vocabulary. Beauty reviews are linked to words such as *hair*, *skin*, *scent*, *makeup*, *shampoo*, and *lotion*. Sports reviews are linked to terms such as *bike*, *fit*, *bag*, *camping*, *comfortable*, and *durable*.

If the bars in the category comparison had remained close to zero, the two categories would have used similar vocabulary. Instead, the long bars show that the subjects and product properties discussed in Beauty and Sports reviews are clearly different.

Together, the vocabulary plots show that positive reviews use broadly similar language across both categories, while negative and product-related vocabulary is more specific to the type of product.

6 Discussion

The results show that the two product categories are similar in their overall sentiment, but different in the language used to describe product experiences.

For RQ1, the sentiment distributions of Beauty & Personal Care and Sports & Outdoors were almost the same. The statistical tests found a difference, but the effect was very small. This means that the difference is unlikely to be important in practice. Therefore, we cannot say that one category is clearly more positive or negative than the other.

For RQ2, both categories followed a similar pattern over time. Sentiment increased until around 2019 and then became lower in the following years. However, there was no clear continuous increase or decrease across the full period. The difference between the two category trends was also very small. This suggests that changes

over time affected both categories in a similar way. The increase seen in 2023 should be treated carefully because the dataset only includes reviews until September.

For RQ3, VADER agreed well with star ratings for positive reviews, but performed much worse for neutral reviews. Many three-star reviews were classified as positive. A possible reason is that three-star reviews often contain both positive and negative comments. For example, a customer may like the product overall but still mention an important problem. In such cases, one sentiment score may not fully represent the complete review.

The high overall agreement was also influenced by the large number of positive reviews. Since most reviews were positive, a method that often predicts positive can still achieve a high accuracy. This is why the results for each sentiment class and the macro-F1 score are important when judging VADER's performance.

Agreement was lower for longer reviews. Longer reviews may discuss several product features and include both positive and negative opinions. This can make the final sentiment harder to summarise with one score. However, the results only show that review length and agreement are related. They do not prove that longer text directly causes lower agreement.

The mismatch analysis showed that VADER more often gave a positive score to a low-rated review than a negative score to a high-rated review. This may happen when a customer uses positive words while explaining why the product did not meet their expectations. It also shows that the written review and the final star rating may focus on different parts of the experience.

RQ4 showed the clearest difference between the categories. Positive reviews in both categories used general words such as *great*, *love*, and *perfect*. Negative reviews were more specific to the product type. Beauty reviews often discussed effectiveness, scent, skin, or hair, while Sports reviews more often mentioned breakage, materials, fit, and durability. This supports earlier work showing that sentiment language can differ between product domains [3].

Overall, the findings show that similar sentiment scores do not mean that customers describe products in the same way. The two categories received similar general evaluations, but the reasons behind those evaluations were different. This suggests that sentiment scores are useful for showing broad patterns, while vocabulary analysis provides more detail about what customers actually liked or disliked.

7 Conclusion

This study compared sentiment and language in Beauty & Personal Care and Sports & Outdoors reviews from Amazon between 2015 and September 2023. We analysed sentiment distributions, changes over time, agreement between VADER and star ratings, and distinctive vocabulary.

The two categories had very similar overall sentiment distributions and followed broadly similar patterns over time. VADER showed moderate agreement with star ratings and worked best for positive reviews, but it had difficulty with neutral reviews and showed lower agreement for longer texts. The vocabulary analysis revealed clearer differences between the categories. Positive reviews mainly used similar general words, while negative reviews described problems that were more specific to each type of product.

Overall, the findings show that sentiment scores can provide a useful overview of customer opinions, but they do not fully explain why customers liked or disliked a product. Combining sentiment analysis with vocabulary analysis gives a more complete view of online reviews. Future work could use manually labelled reviews or models that analyse sentiment for separate product features instead of assigning one score to the complete review.

8 Ethical Considerations

This study uses publicly available Amazon review data and does not involve direct contact with users. The results are shown only for groups, such as product categories and years. We do not try to identify individual reviewers or collect personal information about them. Usernames and other identifying details are not included in the paper.

The results should also be understood with care. VADER may not always understand sarcasm, mixed opinions, or words that are specific to a product category. Its scores should therefore not be treated as a perfect measure of what a reviewer feels. Differences between Beauty and Sports reviews should also not be used to make assumptions about the people who wrote them. Our findings only describe patterns in the selected reviews and may not represent all Amazon users.

9 Use of AI Tools

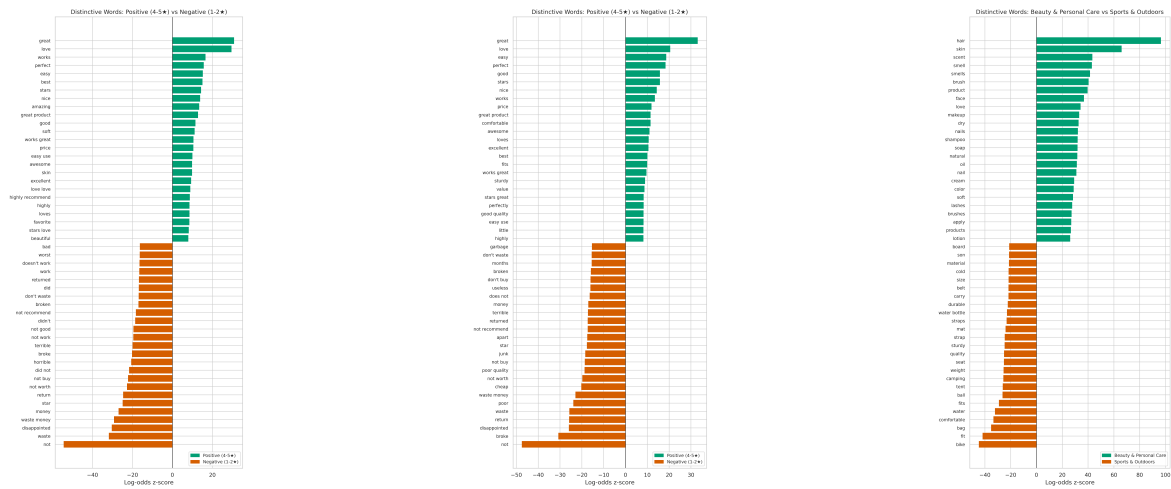
Generative AI tools were used in a limited supporting role to improve the clarity and grammar of the written text and to assist with small parts of the code used to organise and format the retrieved data and the diagrams, thus all outputs were reviewed, adapted, and verified by the authors.

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A RQ4 Vocabulary Figures

Figure 8 presents the complete weighted log-odds results used for RQ4. Figures 8a and 8b compare high-rated and low-rated reviews within each category. Figure 8c compares the vocabulary used in Beauty and Sports reviews directly.



(a) Beauty: high-rated versus low-rated reviews. (b) Sports: high-rated versus low-rated reviews. (c) Direct comparison between the categories.

Figure 8: Distinctive words and two-word phrases identified using weighted log-odds. In (a) and (b), positive values represent high-rated reviews and negative values represent low-rated reviews. In (c), the direction represents the product category rather than sentiment. Longer bars indicate stronger differences between the compared groups.